

TASK 4: TRIGONOMETRIC IDENTITIES

Complete Solutions with Working

Access to HE Diploma in Computing
Mathematics Unit WNI488

PART 1: Using the Identity $\sin^2 x + \cos^2 x = 1$

Section A: Simplify Each Expression

Q1. Simplify $\sin^2 x + \cos^2 x$

$$\sin^2 x + \cos^2 x = 1$$

Answer

1

Q2. Simplify $1 - \sin^2 x$

$$\begin{aligned}\sin^2 x + \cos^2 x &= 1 \\ \Rightarrow \cos^2 x &= 1 - \sin^2 x\end{aligned}$$

Answer

$\cos^2 x$

Q3. Simplify $1 - \cos^2 x$

$$\begin{aligned}\sin^2 x + \cos^2 x &= 1 \\ \Rightarrow \sin^2 x &= 1 - \cos^2 x\end{aligned}$$

Answer

$\sin^2 x$

Q4. Simplify $\frac{\sin^2 x}{\cos^2 x} + 1$

$$\begin{aligned}\frac{\sin^2 x}{\cos^2 x} + 1 &= \frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} \\ &= \frac{\sin^2 x + \cos^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x}\end{aligned}$$

Answer

$$\boxed{\frac{1}{\cos^2 x}} \quad (\text{i.e. } \sec^2 x)$$

Q5. Simplify $\frac{1 - \cos^2 x}{\sin^2 x}$

$$\begin{aligned}\frac{1 - \cos^2 x}{\sin^2 x} &= \frac{\sin^2 x}{\sin^2 x} \\ &= 1\end{aligned}$$

Answer

$$\boxed{1}$$

Section B: Apply the Identity**Q6. Simplify** $5 \sin^2 x + 5 \cos^2 x$

$$\begin{aligned} 5 \sin^2 x + 5 \cos^2 x &= 5(\sin^2 x + \cos^2 x) \\ &= 5(1) \\ &= 5 \end{aligned}$$

Answer**Q7. Simplify** $3 - 3 \sin^2 x$

$$\begin{aligned} 3 - 3 \sin^2 x &= 3(1 - \sin^2 x) \\ &= 3 \cos^2 x \end{aligned}$$

Answer**Q8. Simplify** $2 \cos^2 x - 2 \sin^2 x$

$$\begin{aligned} 2 \cos^2 x - 2 \sin^2 x &= 2(\cos^2 x - \sin^2 x) \\ &= 2((1 - \sin^2 x) - \sin^2 x) \\ &= 2(1 - 2 \sin^2 x) \\ &= 2 - 4 \sin^2 x \end{aligned}$$

Answer or **Q9. Simplify** $\sin^4 x + \cos^4 x$

$$\begin{aligned} (\sin^2 x + \cos^2 x)^2 &= \sin^4 x + 2 \sin^2 x \cos^2 x + \cos^4 x \\ 1 &= \sin^4 x + 2 \sin^2 x \cos^2 x + \cos^4 x \\ \Rightarrow \sin^4 x + \cos^4 x &= 1 - 2 \sin^2 x \cos^2 x \end{aligned}$$

Answer**Q10. Simplify** $\frac{1 - \cos^2 x}{1 + \cos^2 x}$

$$\frac{1 - \cos^2 x}{1 + \cos^2 x} = \frac{\sin^2 x}{1 + \cos^2 x}$$

Answer

$$\frac{\sin^2 x}{1 + \cos^2 x}$$

Section C: Challenge Questions (Part 1)

Q11. Simplify completely $\frac{\sin^2 x + \cos^2 x}{1 - \sin^2 x}$

$$\begin{aligned}\frac{\sin^2 x + \cos^2 x}{1 - \sin^2 x} &= \frac{1}{1 - \sin^2 x} \\ &= \frac{1}{\cos^2 x}\end{aligned}$$

Answer

$$\frac{1}{\cos^2 x} \quad (\text{i.e. } \sec^2 x)$$

Q12. Simplify $\sin^2 x(1 - \sin^2 x) + \cos^2 x(1 - \cos^2 x)$

$$\begin{aligned}\sin^2 x(1 - \sin^2 x) + \cos^2 x(1 - \cos^2 x) &= \sin^2 x \cos^2 x + \cos^2 x \sin^2 x \\ &= 2 \sin^2 x \cos^2 x\end{aligned}$$

Answer

$$2 \sin^2 x \cos^2 x$$

Q13. Simplify $\frac{\sin^2 x}{1 - \sin^2 x}$

$$\frac{\sin^2 x}{1 - \sin^2 x} = \frac{\sin^2 x}{\cos^2 x}$$

Answer

$$\frac{\sin^2 x}{\cos^2 x} \quad (\text{i.e. } \tan^2 x)$$

Q14. Simplify $\sin^2 x + \sin^2 x \cos^2 x$

$$\sin^2 x + \sin^2 x \cos^2 x = \sin^2 x(1 + \cos^2 x)$$

Answer

$$\sin^2 x(1 + \cos^2 x)$$

Q15. Simplify $\frac{1 - 2 \cos^2 x}{\sin^2 x}$

$$\begin{aligned}\frac{1 - 2 \cos^2 x}{\sin^2 x} &= \frac{1 - 2(1 - \sin^2 x)}{\sin^2 x} \\ &= \frac{-1 + 2 \sin^2 x}{\sin^2 x} \\ &= 2 - \frac{1}{\sin^2 x}\end{aligned}$$

Answer

$$\boxed{2 - \frac{1}{\sin^2 x}} \quad (\text{i.e. } 2 - \csc^2 x)$$

PART 2: Using the Identity $\tan x = \frac{\sin x}{\cos x}$

Section A: Simplify Each Expression

Q1. Simplify $\frac{\sin x}{\cos x}$

$$\frac{\sin x}{\cos x} = \tan x$$

Answer

$$\tan x$$

Q2. Simplify $\frac{\sin x}{\frac{1}{\cos x}}$

$$\begin{aligned} \frac{\sin x}{\frac{1}{\cos x}} &= \sin x \times \cos x \\ &= \sin x \cos x \end{aligned}$$

Answer

$$\sin x \cos x$$

Q3. Simplify $\frac{\cos x}{\sin x}$

$$\begin{aligned} \frac{\cos x}{\sin x} &= \frac{1}{\frac{\sin x}{\cos x}} \\ &= \frac{1}{\tan x} \end{aligned}$$

Answer

$$\frac{1}{\tan x} \quad (\text{i.e. } \cot x)$$

Q4. Simplify $\frac{\sin^2 x}{\cos^2 x}$

$$\begin{aligned} \frac{\sin^2 x}{\cos^2 x} &= \left(\frac{\sin x}{\cos x} \right)^2 \\ &= \tan^2 x \end{aligned}$$

Answer

$$\tan^2 x$$

Q5. Simplify $\sin x \times \frac{1}{\cos x}$

$$\begin{aligned}\sin x \times \frac{1}{\cos x} &= \frac{\sin x}{\cos x} \\ &= \tan x\end{aligned}$$

Answer

$$\tan x$$

Section B: Apply the Identity

Q6. Simplify $\frac{1 + \tan^2 x}{1 + \frac{\sin^2 x}{\cos^2 x}}$

$$1 + \tan^2 x = 1 + \frac{\sin^2 x}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$1 + \frac{\sin^2 x}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$\frac{1 + \tan^2 x}{1 + \frac{\sin^2 x}{\cos^2 x}} = \frac{\frac{1}{\cos^2 x}}{\frac{1}{\cos^2 x}} = 1$$

Answer

Q7. Simplify $\frac{\tan x}{\sin x}$

$$\frac{\tan x}{\sin x} = \frac{\frac{\sin x}{\cos x}}{\sin x}$$

$$= \frac{1}{\cos x}$$

Answer (i.e. sec x)

Q8. Simplify $\tan x \times \cos x$

$$\tan x \times \cos x = \frac{\sin x}{\cos x} \times \cos x$$

$$= \sin x$$

Answer

Q9. Simplify $\frac{\tan x}{\cos x}$

$$\frac{\tan x}{\cos x} = \frac{\frac{\sin x}{\cos x}}{\cos x}$$

$$= \frac{\sin x}{\cos^2 x}$$

Answer

$$\frac{\sin x}{\cos^2 x}$$

Q10. Simplify $\frac{\sin x(1 + \tan^2 x)}{\cos x}$

$$\begin{aligned} 1 + \tan^2 x &= \sec^2 x = \frac{1}{\cos^2 x} \\ \frac{\sin x(1 + \tan^2 x)}{\cos x} &= \frac{\sin x \left(\frac{1}{\cos^2 x} \right)}{\cos x} \\ &= \frac{\sin x}{\cos^3 x} \end{aligned}$$

Answer

$$\frac{\sin x}{\cos^3 x}$$

Section C: Challenge Questions (Part 2)

Q11. Simplify $\frac{\tan^2 x}{1 + \tan^2 x}$

$$\begin{aligned}\frac{\tan^2 x}{1 + \tan^2 x} &= \frac{\frac{\sin^2 x}{\cos^2 x}}{1 + \tan^2 x} \\ \text{but } 1 + \tan^2 x &= \sec^2 x = \frac{1}{\cos^2 x} \\ \Rightarrow \frac{\tan^2 x}{1 + \tan^2 x} &= \frac{\frac{\sin^2 x}{\cos^2 x}}{\frac{1}{\cos^2 x}} \\ &= \left(\frac{\sin^2 x}{\cos^2 x} \right) (\cos^2 x) \\ &= \sin^2 x\end{aligned}$$

Answer

$$\boxed{\sin^2 x}$$

Q12. Simplify $\frac{\sin x + \tan x}{\cos x}$

$$\begin{aligned}\frac{\sin x + \tan x}{\cos x} &= \frac{\sin x + \frac{\sin x}{\cos x}}{\cos x} \\ &= \frac{\frac{\sin x \cos x + \sin x}{\cos x}}{\cos x} \\ &= \frac{\sin x(\cos x + 1)}{\cos^2 x}\end{aligned}$$

Answer

$$\boxed{\frac{\sin x(1 + \cos x)}{\cos^2 x}}$$

Q13. Simplify $\frac{\sin x(1 - \tan^2 x)}{\cos x}$

$$\begin{aligned}1 - \tan^2 x &= 1 - \frac{\sin^2 x}{\cos^2 x} = \frac{\cos^2 x - \sin^2 x}{\cos^2 x} \\ \frac{\sin x(1 - \tan^2 x)}{\cos x} &= \frac{\sin x(\cos^2 x - \sin^2 x)}{\cos^3 x}\end{aligned}$$

Answer

$$\frac{\sin x(\cos^2 x - \sin^2 x)}{\cos^3 x}$$

Q14. Simplify $\frac{\tan x}{1 - \tan^2 x}$

$$\begin{aligned} \frac{\tan x}{1 - \tan^2 x} &= \frac{\frac{\sin x}{\cos x}}{1 - \frac{\sin^2 x}{\cos^2 x}} \\ &= \frac{\frac{\sin x}{\cos x}}{\frac{\cos^2 x - \sin^2 x}{\cos^2 x}} \\ &= \frac{\sin x}{\cos x} \cdot \frac{\cos^2 x}{\cos^2 x - \sin^2 x} \\ &= \frac{\sin x \cos x}{\cos^2 x - \sin^2 x} \end{aligned}$$

Answer

$$\frac{\sin x \cos x}{\cos^2 x - \sin^2 x}$$

Q15. Simplify $\frac{\sin^3 x}{\cos^3 x}$

$$\begin{aligned} \frac{\sin^3 x}{\cos^3 x} &= \left(\frac{\sin x}{\cos x} \right)^3 \\ &= \tan^3 x \end{aligned}$$

Answer

$$\tan^3 x$$

Key Identities Used

Identity	Formula
Pythagorean Identity	$\sin^2 x + \cos^2 x = 1$
Tangent Definition	$\tan x = \frac{\sin x}{\cos x}$
Secant	$\sec x = \frac{1}{\cos x}$
Tangent–Secant	$1 + \tan^2 x = \sec^2 x = \frac{1}{\cos^2 x}$